

Programmable Peripheral Interface

- Methods of parallel data transfer
 - Simple input/output
 - Simple strobe I/O
 - Single handshake I/O
 - Double handshake I/O

Simple I/O

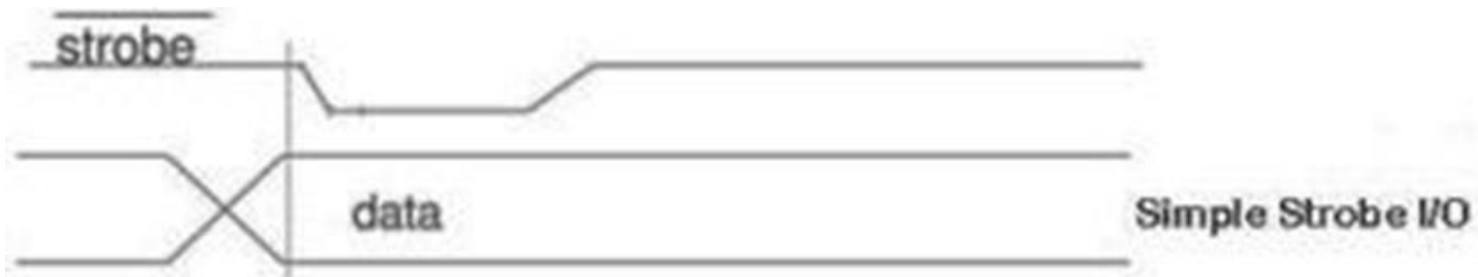
- This data transfer method is used when the I/O devices need no communication before the data transfer.
- The crossed lines on the wave form represent the time at which a new data byte becomes valid on the output lines of the port.



simple
(no handshake)

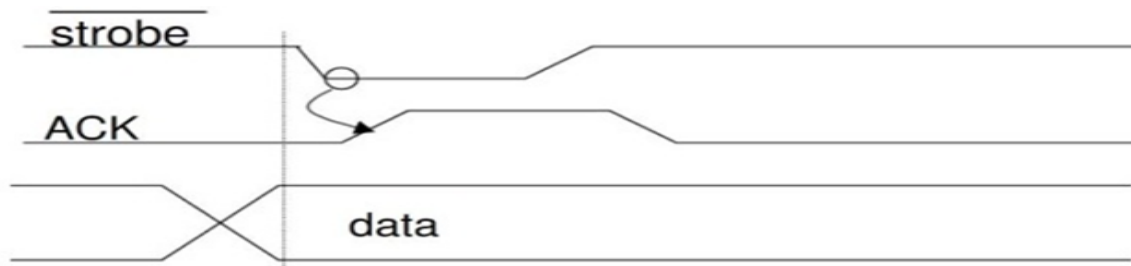
Simple Strobe I/O

- The sending device, such as a keyboard, outputs parallel data on the data lines, and then outputs an strobe' signal to let the receiving device know that valid data is present.



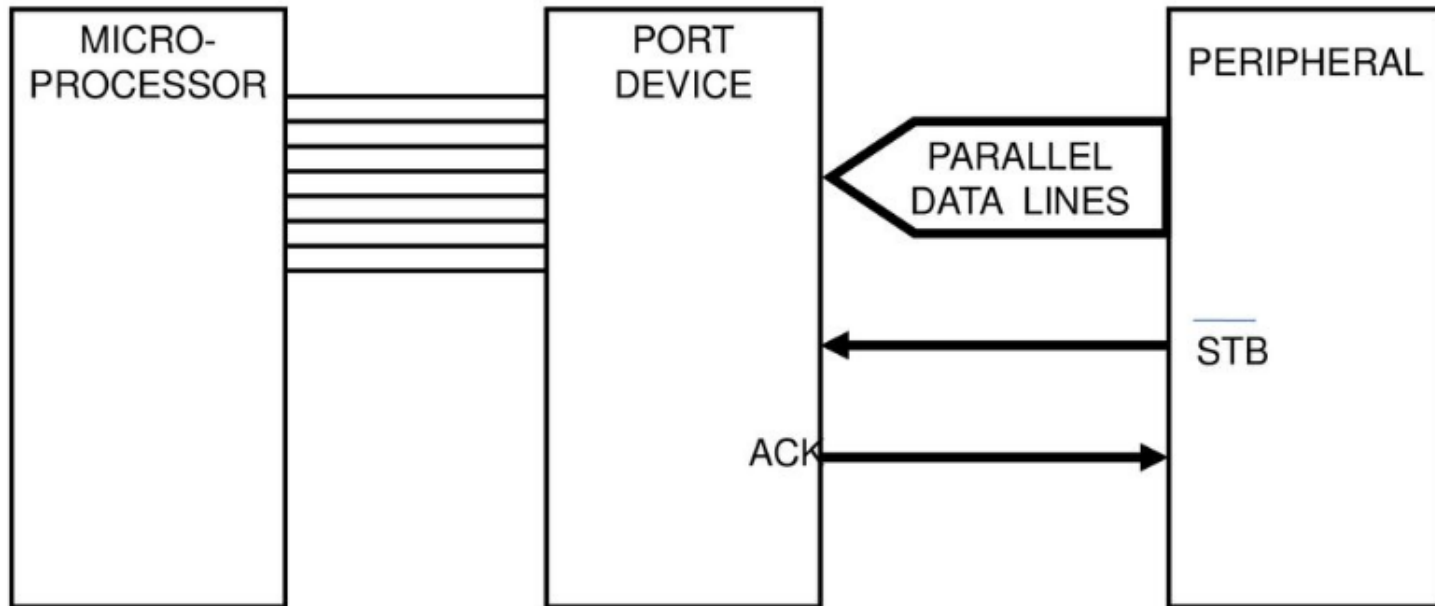
Single handshake I/O

- Peripheral device outputs some parallel data and send STB'signal to MP.
- MP detects STB' signal and sends ACK to peripheral to indicate data has been read and peripheral can sent next byte of data.
- Handshaking ensures that peripheral device does not send data until receiving system indicates with an ACK that it is ready to receive next byte.

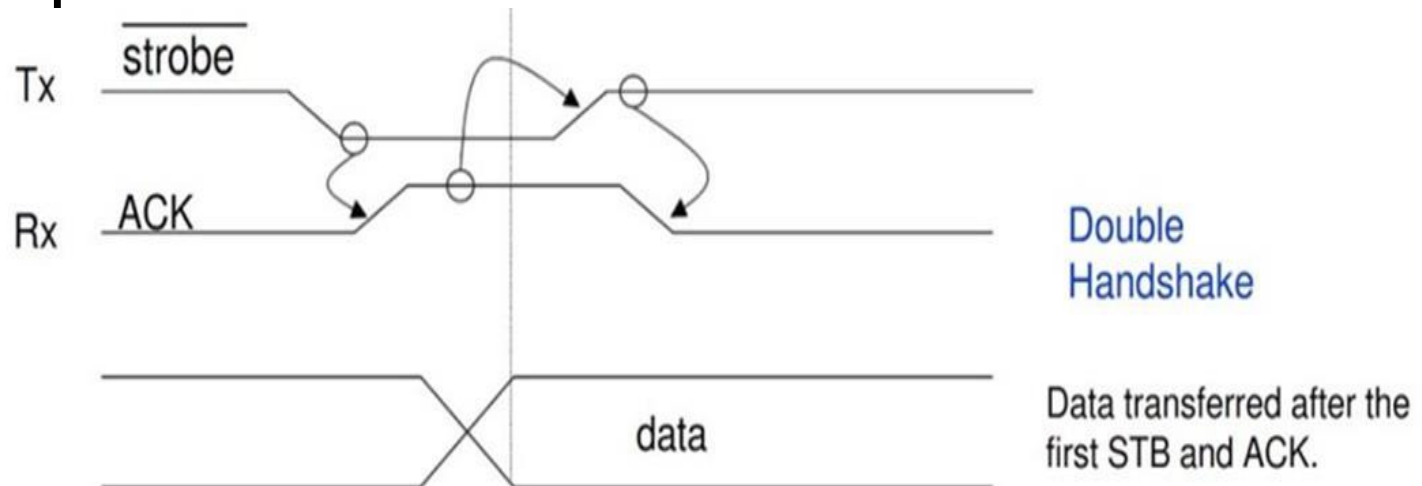


Single
Handshake

HANDSHAKE INPUT DATA TRANSFER SIGNAL DIRECTIONS



- The sender asserts $\overline{STB'}$ (low) and the receiver raises its ACK.
- The external device then send data byte and raises its $\overline{STB'}$ and receiver reads the data and drop ACK.



Differences between Isolated I/O and Memory Mapped I/O:

Isolated I/O	Memory Mapped I/O
Isolated I/O uses separate memory space.	Memory mapped I/O uses memory from the main memory.
Limited instructions can be used. Those are IN, OUT.	Any instruction which references to memory can be used. (MOV, AND XCHG, SUB)
The memory address space is not affected	Part of the memory address space is lost
The addresses for Isolated I/O devices are called ports.	Memory mapped I/O devices are treated as memory locations on the memory map

INPUT-OUTPUT INSTRUCTIONS

- IN – IN Accumulator, Port
- The IN instruction copies data from a port to the AL or AX register.
- If an 8-bit port is read, the data will go to AL.
- If a 16-bit port is read, the data will go to AX.
- The IN instruction has two possible formats, fixed port and variable port.
- For fixed port type, the 8-bit address of a port is specified directly in the instruction.

Ex: IN AL, 0C8H ; Input a byte from port 0C8H to AL
 IN AX, 34H ; Input a word from port 34H to AX

- For the variable-port form of the IN instruction, the port address is loaded into the DX register before the IN instruction.
- Since DX is a 16-bit register, the port address can be any number between 0000H and FFFFH.
- The IN instruction does not change any flag.

Ex: MOV DX, 0FF78H

IN AL, DX

IN AX, DX

OUT – OUT Port, Accumulator

- The OUT instruction copies a byte from AL or a word from AX to the specified port.
- The OUT instruction has two possible forms, fixed port and variable port.
- For the fixed port form, the 8-bit port address is specified directly in the instruction. With this form, any one of 256 possible ports can be addressed.
- Ex: OUT 3BH, AL
OUT 2CH, AX

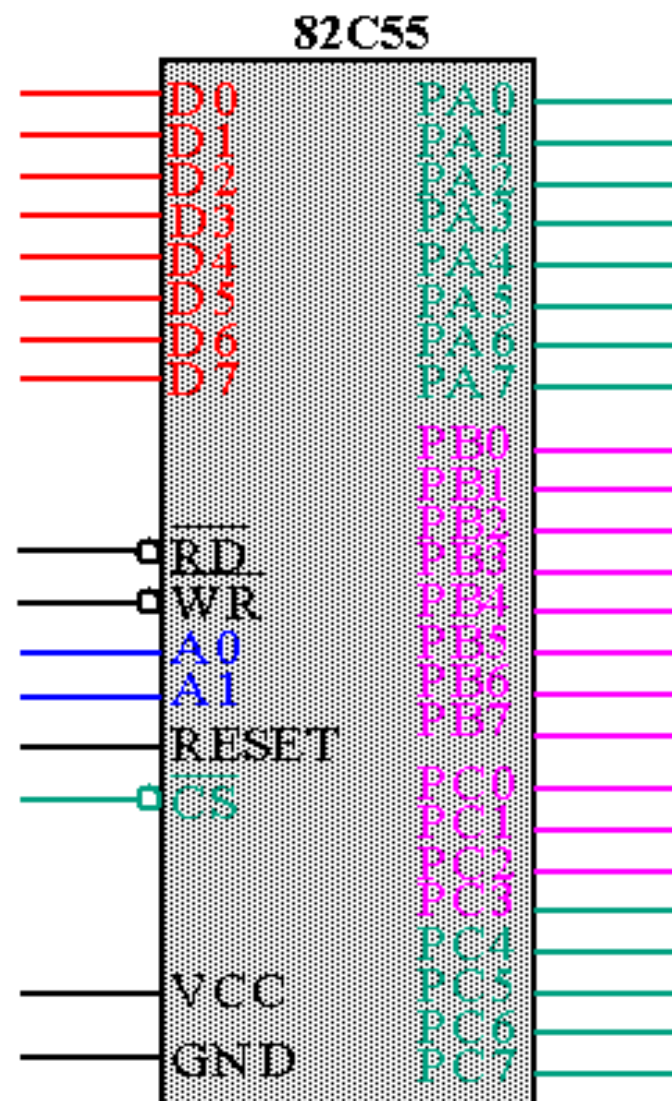
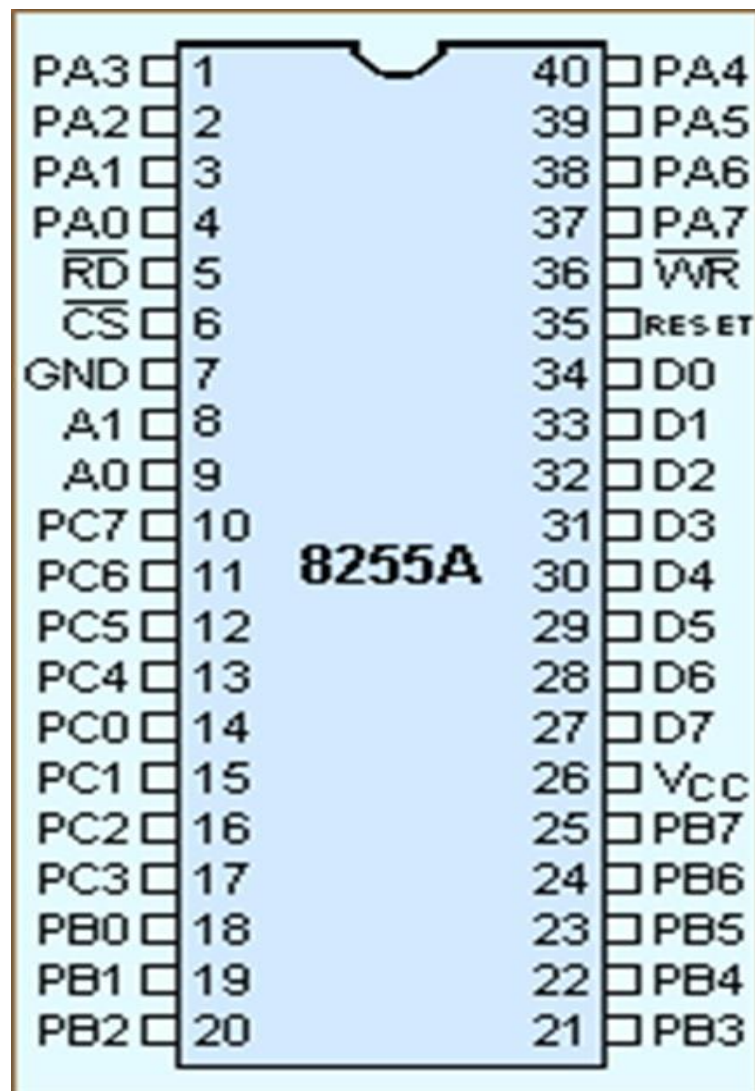
- For variable port form of the OUT instruction, the content of AL or AX will be copied to the port at an address contained in DX.
- Therefore, the DX register must be loaded with the desired port address before this form of the OUT instruction is used.
- `MOV DX, 0FFF8H`
`OUT DX, AL`
`OUT DX, AX`
The OUT instruction does not affect any flag

Need of 8255

- The parallel port device 8255 has been designed to automatically manage the handshake operation.
- It can be programmed to automatically receive an Strobe signal from a peripheral, send an int signal to the processor and send the ACK signal back to the peripheral at proper times

- The Intel 82C55A is a general purpose programmable I/O device which may be used with many different microprocessors.
- It is used to interface to the keyboard and a parallel printer port in PCs (usually as part of an integrated chipset).
- Requires insertion of wait states if used with a microprocessor using higher than an 8 MHz clock.

- There are 24 I/O pins which may be individually programmed in 2 groups of 12 and used in 3 major modes of operation.
- The high performance and industry standard configuration of the 82C55A make it compatible with the 8086.



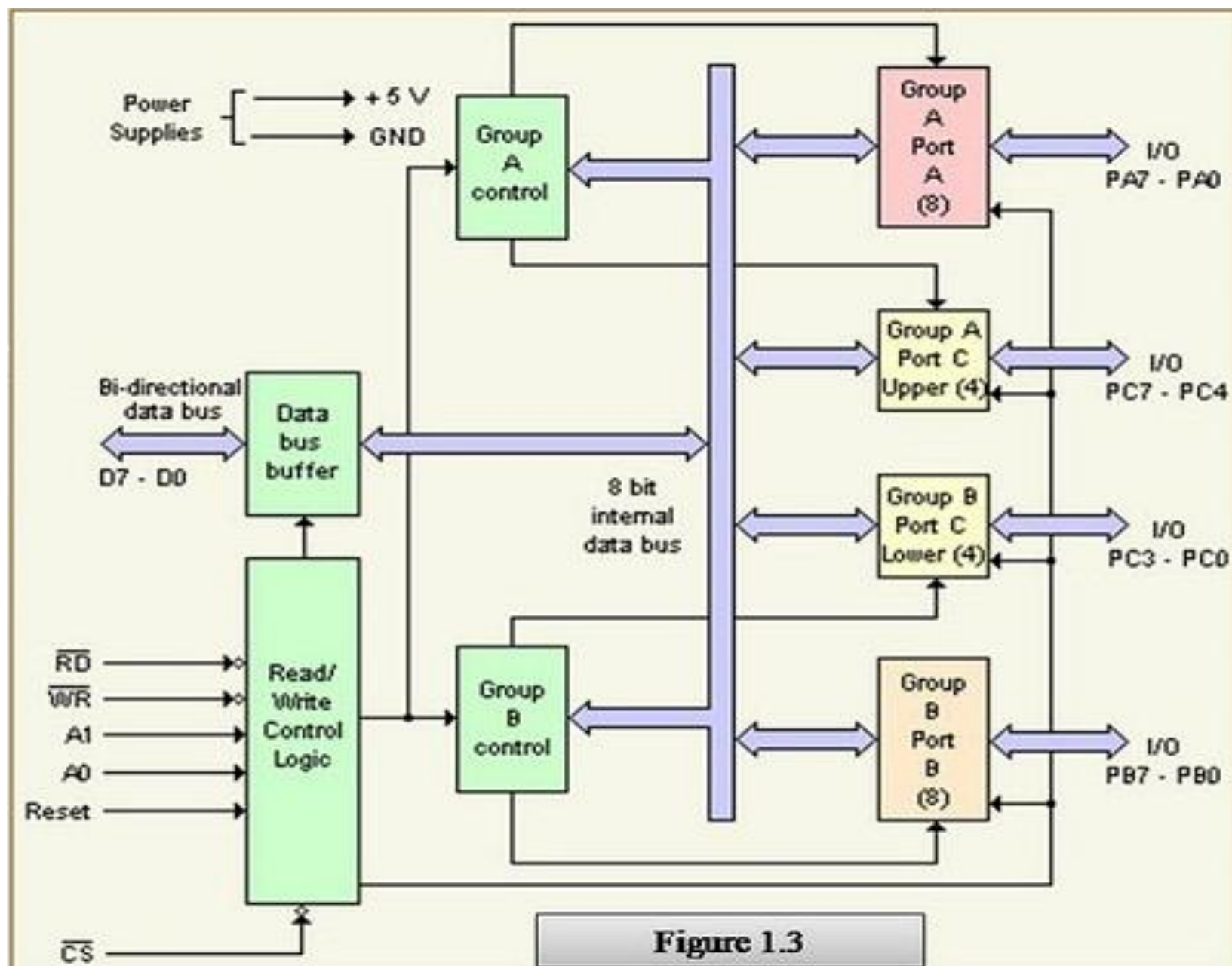


Figure 1.3

- Data Bus Buffer
- It is a tri-state 8-bit buffer, which is used to interface the microprocessor to the system data bus.
- Data is transmitted or received by the buffer as per the instructions by the CPU. Control words and status information is also transferred using this bus.

Read/Write Control Logic

- The function of this block is to manage all of the internal and external transfers of both Data and Control or Status words.
- **(CS) Chip Select:** A "low" on this input pin enables the communication between the 8255 and the CPU.
- **(RD) Read:** A "low" on this input pin enables 8255 to send the data or status information to the CPU on the data bus. It allows the CPU to "read from" the 8255.
- **(WR) Write:** A "low" on this input pin enables the CPU to write data or control words into the 8255.

- **(A0 and A1)** These input signals, control the selection of one of the three ports or the control word register.

A1	A0	SELECTION
0	0	PORT A
0	1	PORT B
1	0	PORT C
1	1	CONTROL

- **(RESET)** Reset. A "high" on this input initializes the control register to 9Bh and all ports (A, B, C) are set to the input mode.

Group A and Group B Controls

- Each of the Control blocks (Group A and Group B) accepts "commands" from the CPU as "control word" and configure the ports (Port A, Port B and Port C) accordingly.
- **Port A and upper 4 bits of Port C are controlled by Group A**
- **Port B and lower 4 bits of Port C are controlled by Group B**

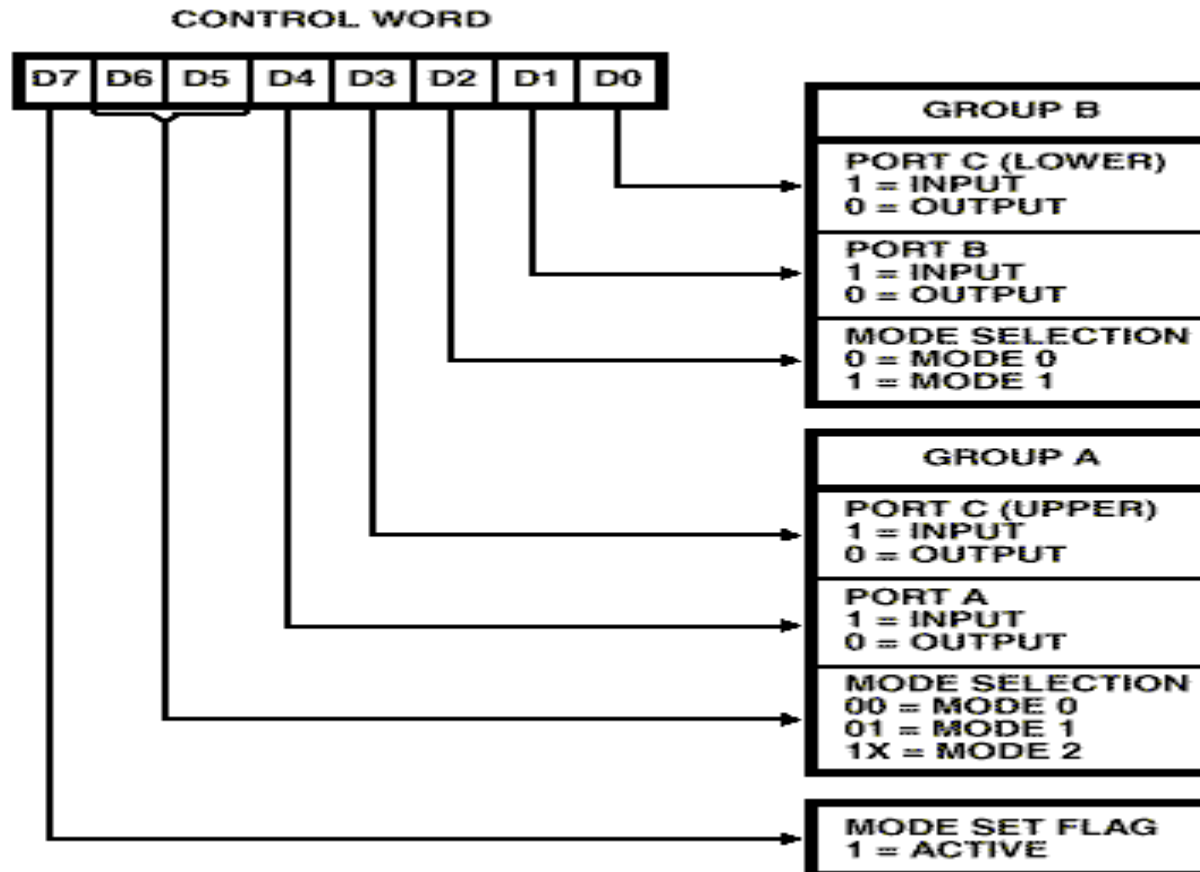
Ports A, B, and C

- The 8255 has three 8 bit I/O ports and each one can be connected to the physical lines of an external device. All can be configured to a wide variety of functional characteristics by the system software These ports are labeled as PA0-PA7 (PortA), PB0-PB7 (PortB) and PC0-PC7 (PortC).
- **GND (Ground) and Vcc**

Programming 8255A

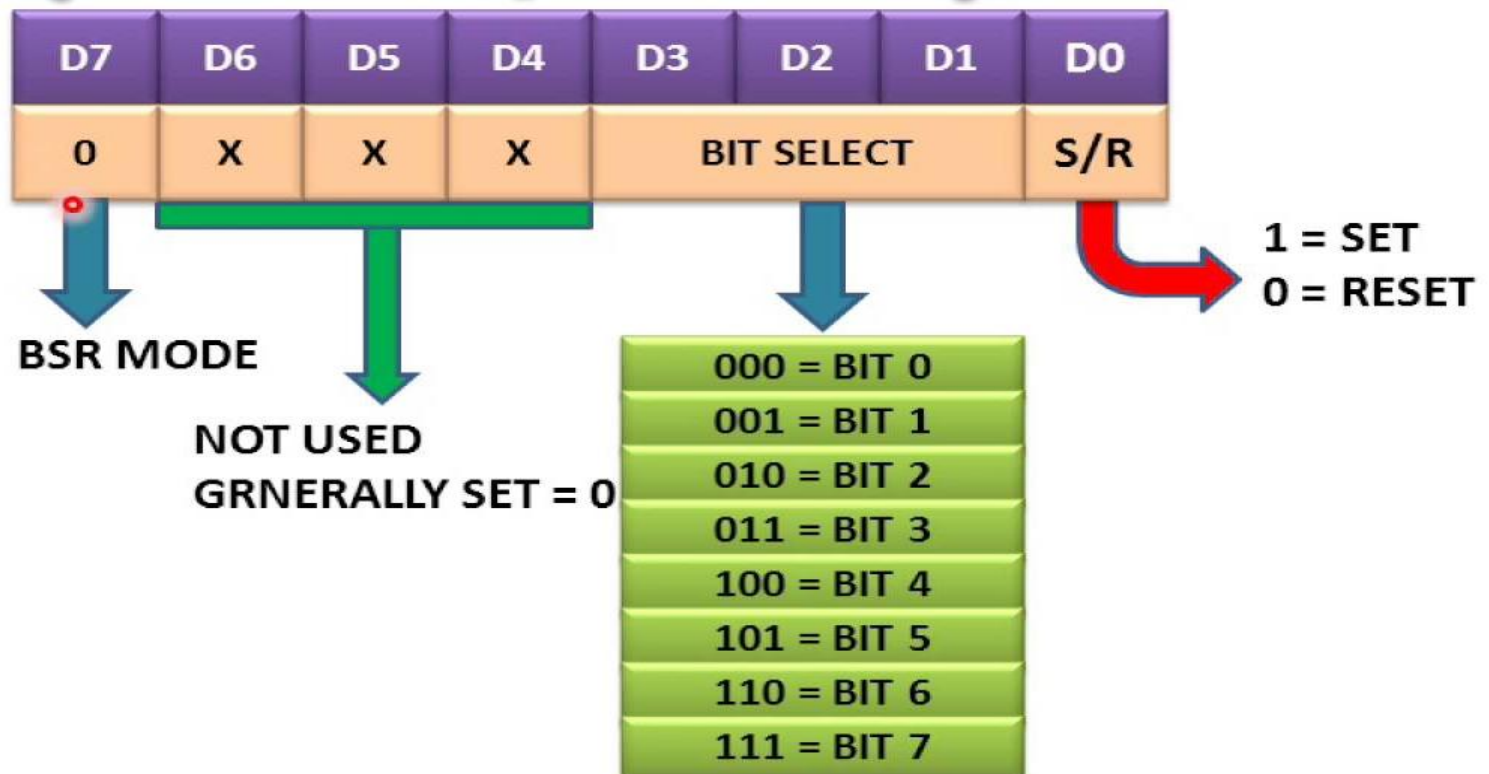
- **Modes of Operation**
- 8255 can be configured in two modes
 - BSR (Bit Set Reset) Mode
 - I/O (Input-Output) Mode: Mode 0, Mode 1 and Mode 2
- Modes are configured by Control Word

- A Control Word is an 8-bit data that stored in control register. Control Words are two types:
(a) BSR Control Word (b) Mode definition Control Word



Bit Set Reset Mode

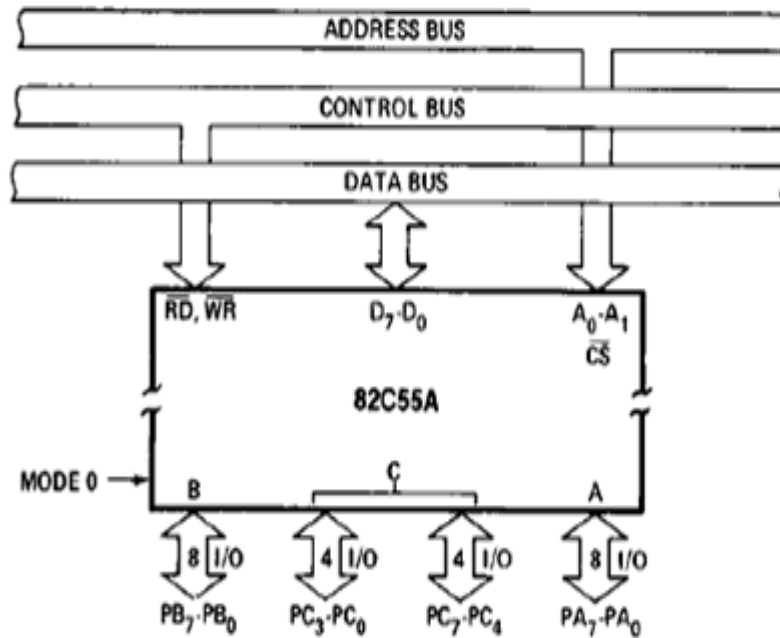
- If bit 7 of control word is a logic 0 then 8255 will be configured as BSR (Bit Set Reset) mode.
- In this mode we can set or reset the pins of port C



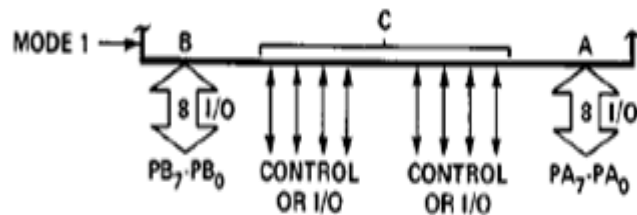
Working Modes Of 8255

- There are 3 modes of operation
 - I. Mode 0 -- Basic Input/Output
 - II. Mode 1 – Strobed Input/Output
 - III. Mode 2 – Bidirectional Bus
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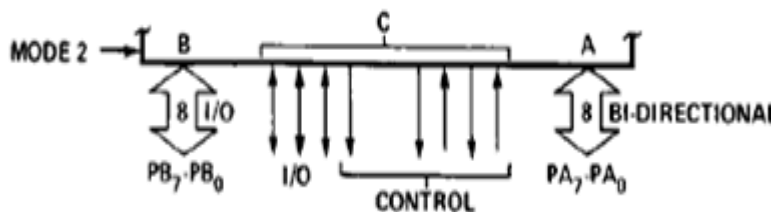
Basic Mode Definitions and Bus Int



Mode 0
Basic I/O



Mode 1
Strobe I/O

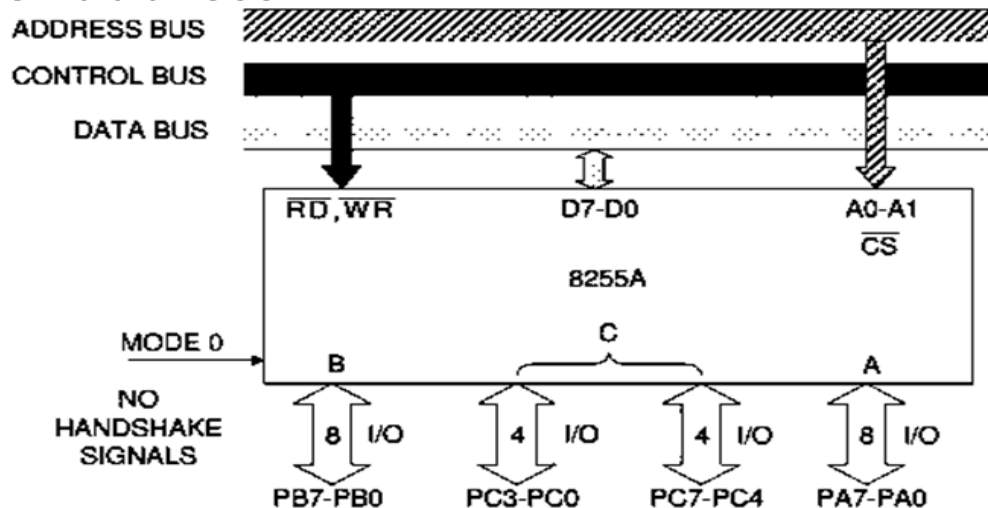


Mode 2
Bi-Dir Bus

I/O Mode (Configured by Mode definition CR)

Mode 0

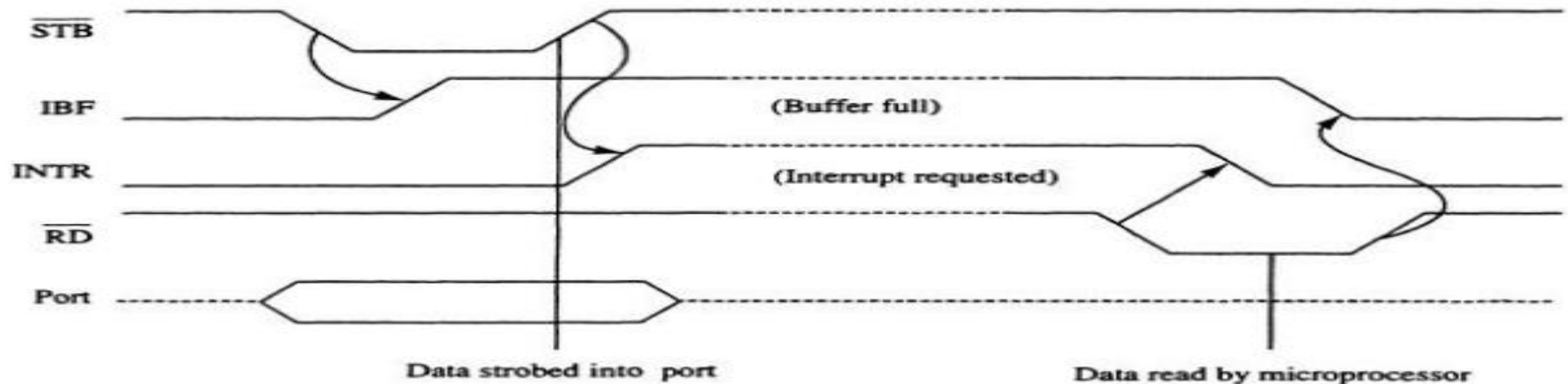
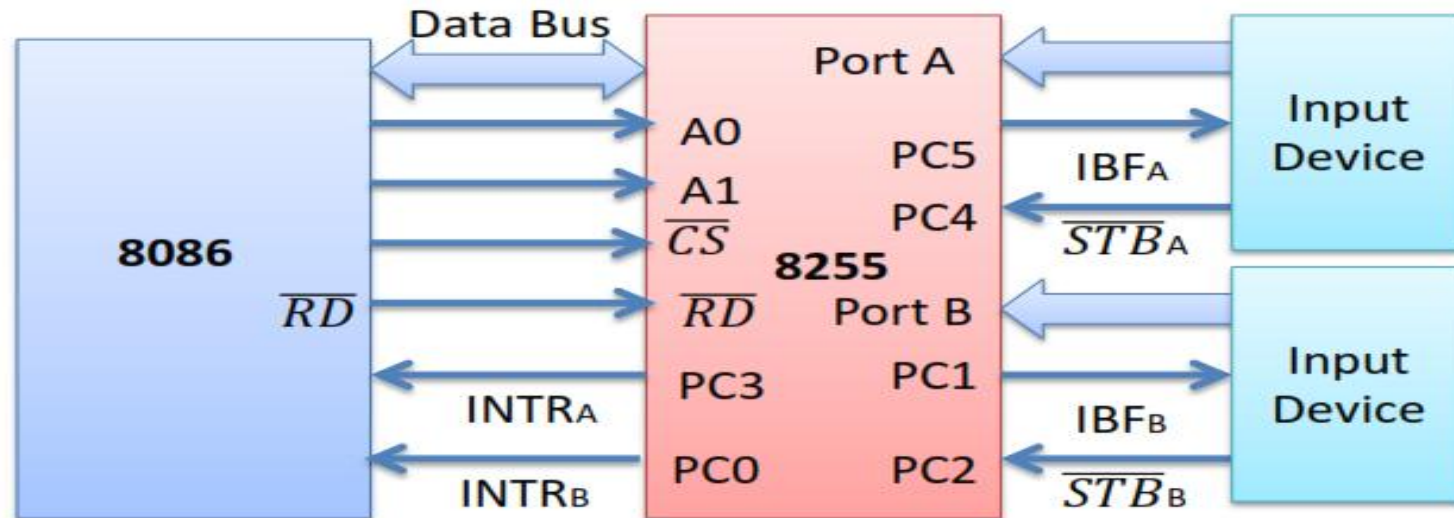
- Port A works as simple input or output **without handshaking.**
- Port B works as simple input or output without handshaking.
- Port C can be used together as an additional 8 bit port or they can be used individually as two 4-bit ports.
- When used as outputs, the Port C lines can be individually set or reset by sending a special control word to the control register address.



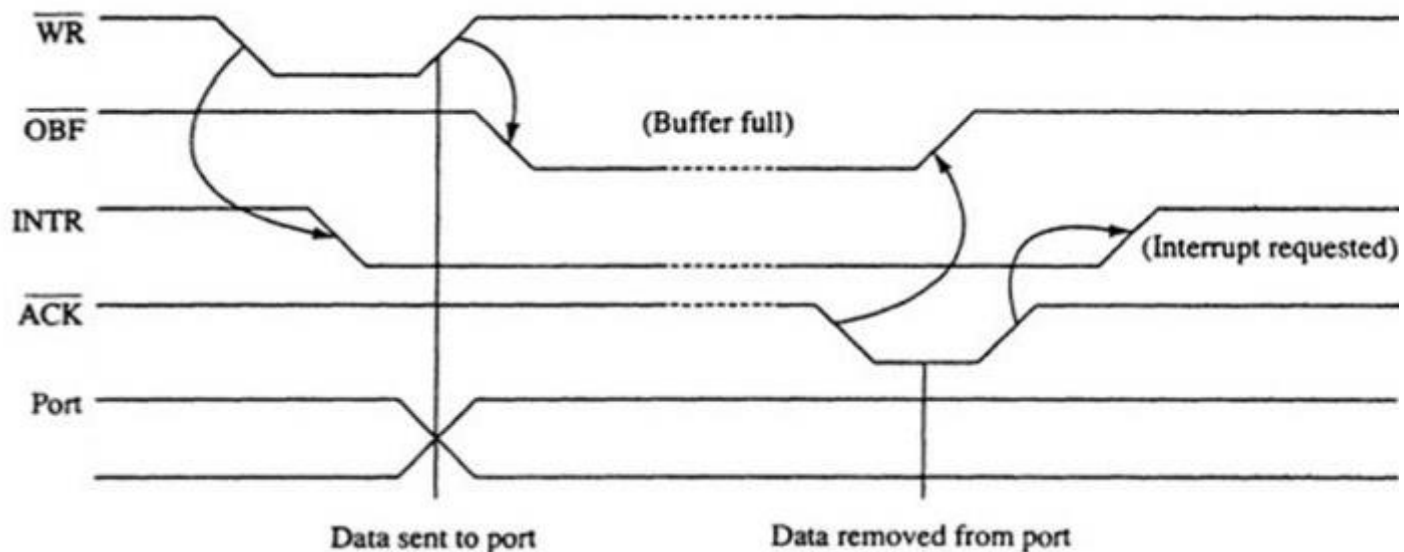
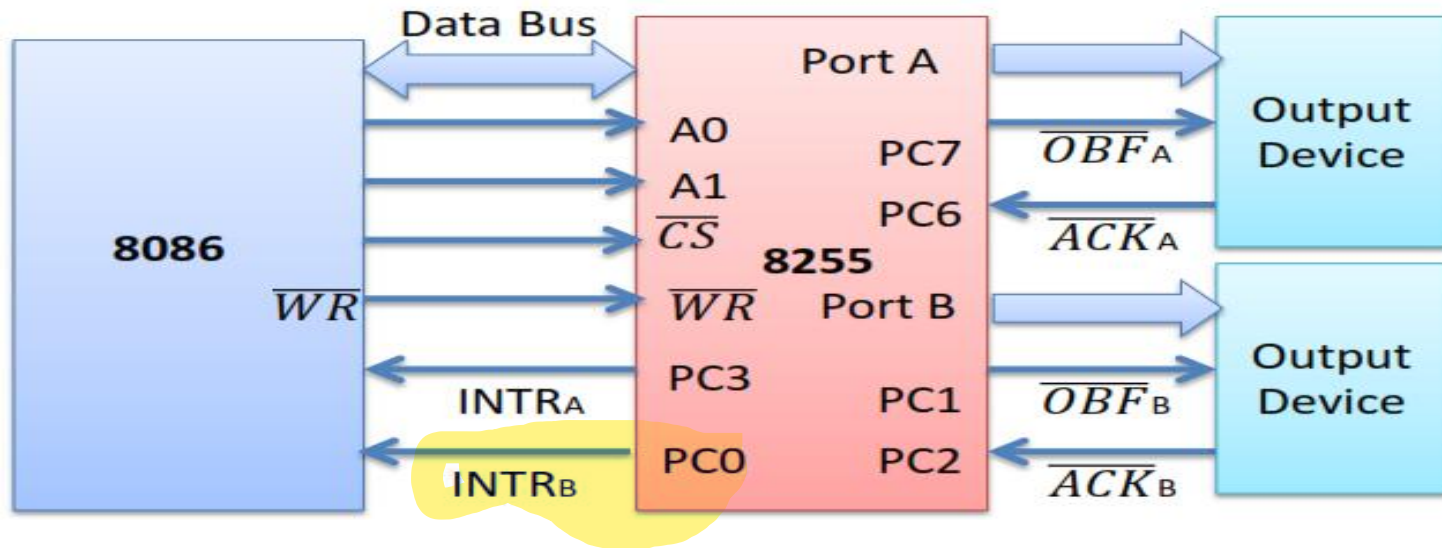
Mode 1

- Used for handshake input/output operation.
- Port B is initialized in mode 1 for either input or output, Pins PC0, PC1 and PC2 function as handshake lines.
- Port A can also be configured as input or output in mode 1. But handshake signal pins are not same for input and output mode as like Port B.
- If port A is initialized in mode 1 as handshake input port, then pins PC3, PC4 and PC5 function as handshake signals. (PC6 and PC7 are available for using as input lines or output lines)
- If port A is initialized as handshake output port, then PC3, PC6 and PC7 function as handshake signals. (PC4 and PC5 are available for using as input or output lines)

Port A and Port B configured as Input Port



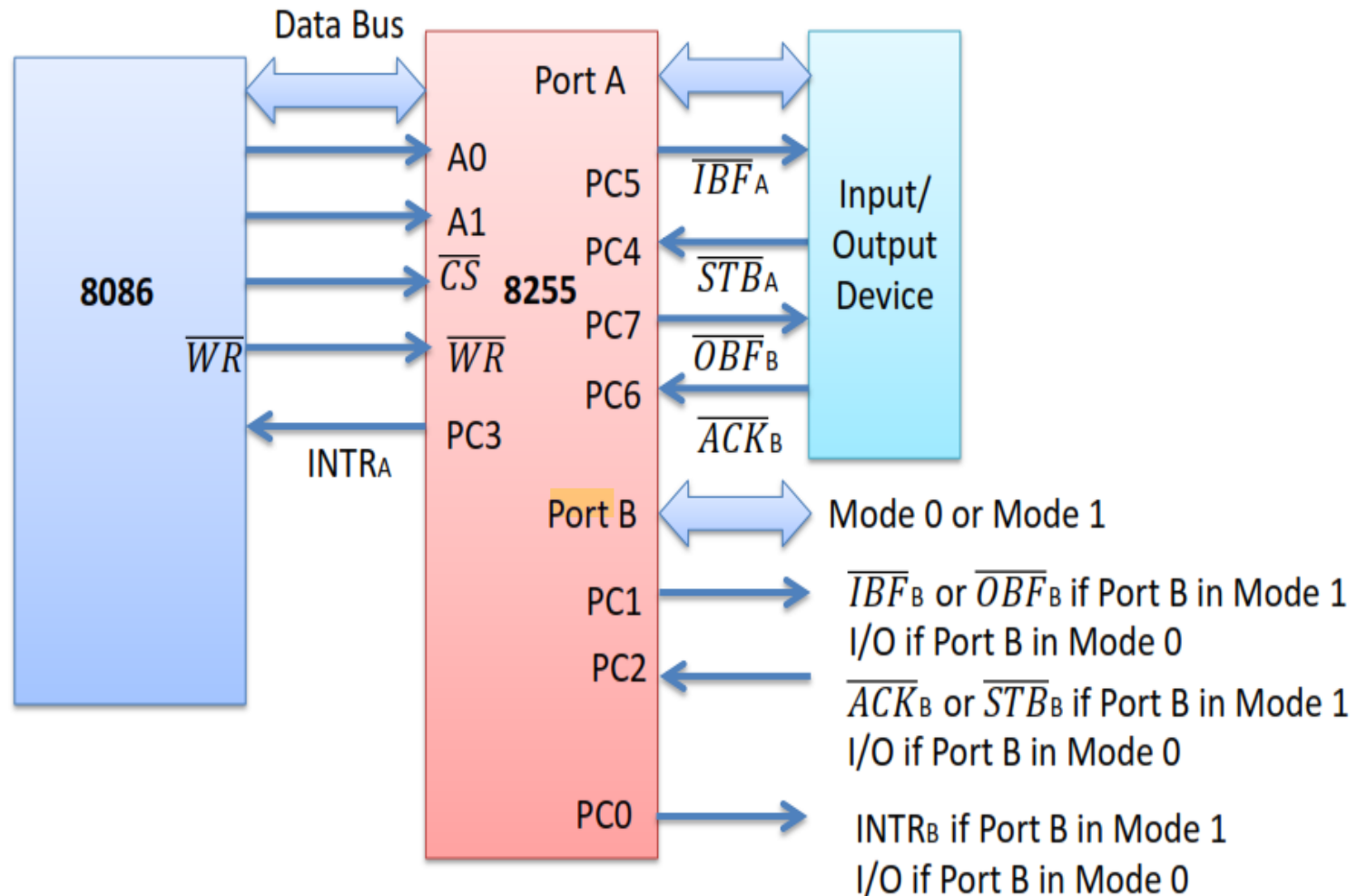
Port A and Port B configured as Output Port



Mode 2

- Only port A can be initialized in mode 2.
- In mode 2, port A can be used for “bi-directional handshake” data transfer i.e. data can be input or output on the same eight lines.
- Pins PC3, PC4, PC5, PC6, PC7 used as handshake lines for port A.
- Port B is operating in either mode 0 or mode 1.
- If port B is in mode 0, then PC0, PC1 and PC2 used for I/O.
- If port B is in mode 1, then PC0, PC1 and PC2 used as handshake lines.

Mode 2 Handshake Signals



Summary of 8255 modes

Mode	8- Bit Port A	8-Bit Port B	Handshake lines for mentioned Mode	Other Port C pins available for I/O
0	I/O	I/O	N/A	PC7- PC4 and PC3- PC0
1	Input	I/O	PC3, PC4, PC5 (Port A) PC0, PC1, PC2 (Port B)	PC6, PC7
1	Output	I/O	PC3, PC6, PC7 (Port A) PC0, PC1, PC2 (Port B)	PC4, PC5
2	Bi-Directional	N/A	PC3, PC4, PC5, PC6, PC7	PC0, PC1, PC2 (If Port B in Mode 0)

- Write a control word to configure port A as input port in mode 0 and port B in mode 1 as output port.

D7	D6	D5	D4	D3	D2	D1	D0
1	0	0	1	0	1	0	0

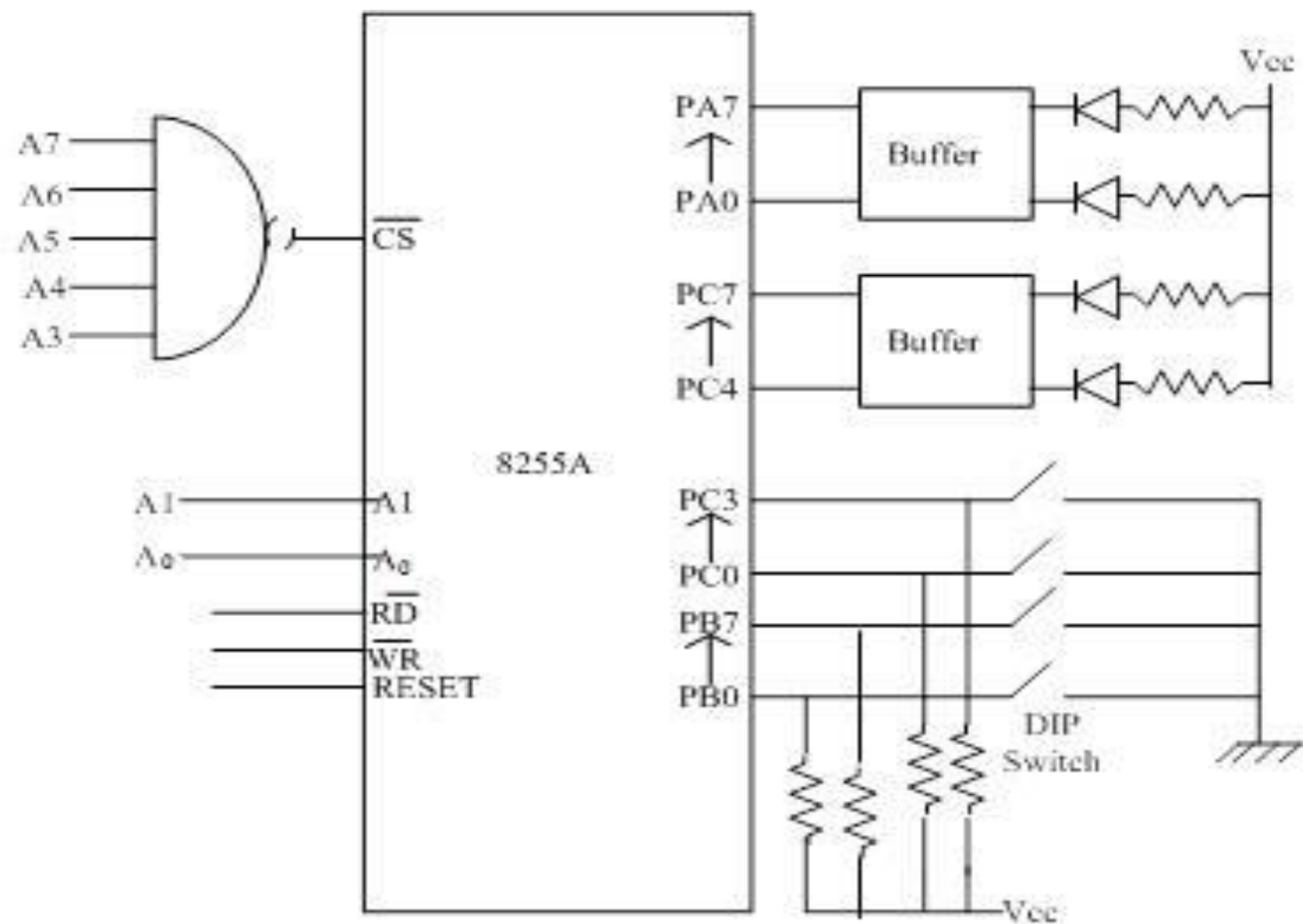
- Write an 8086 assembly language procedure to read an ASCII character from a keyboard via PORT A of an 8255 PPI when PORT C bit PC4 is strobed low. Assume a base address of 20H.

```
PORTA      EQU    20H
PORTC      EQU    24H
CONTROL    EQU    26H
```

```
MOV AL, 0B8H ; 1011 1000
OUT CONTROL, AL ;
READ1: IN AL, PORTC
TEST AL, 10H ; 0001 0000
JNZ READ1
IN AL, PORTA
HLT
```

Figure shows an 8255A interfaced with 8086 microprocessor. Perform the following-

- (a) Identify the Port Address.
- (b) Identify the Mode 0 control word to configure Port A and Port CU as output ports and Port B and Port CL as input ports.
- (c) Write a program to read the DIP switches and display the reading from Port B at Port A, and from Port CL at Port CU



- 00F8H (Port A)
- 00F9H (Port B)
- 00FAH (Port C)
- 00FBH (CWR)

Control Word=83H

D7	D6	D5	D4	D3	D2	D1	D0
1	0	0	0	0	0	1	1
Mod Set	Port A in Mode 0		Port A I/O	Port C upper o/p	Port B in Mode 0	Port B input	Port C lower i/p

Program

```
PPICR EQU 00FBH  
PPIC EQU 00FAH  
PPIB EQU 00F9H  
PPIA EQU 00F8H
```

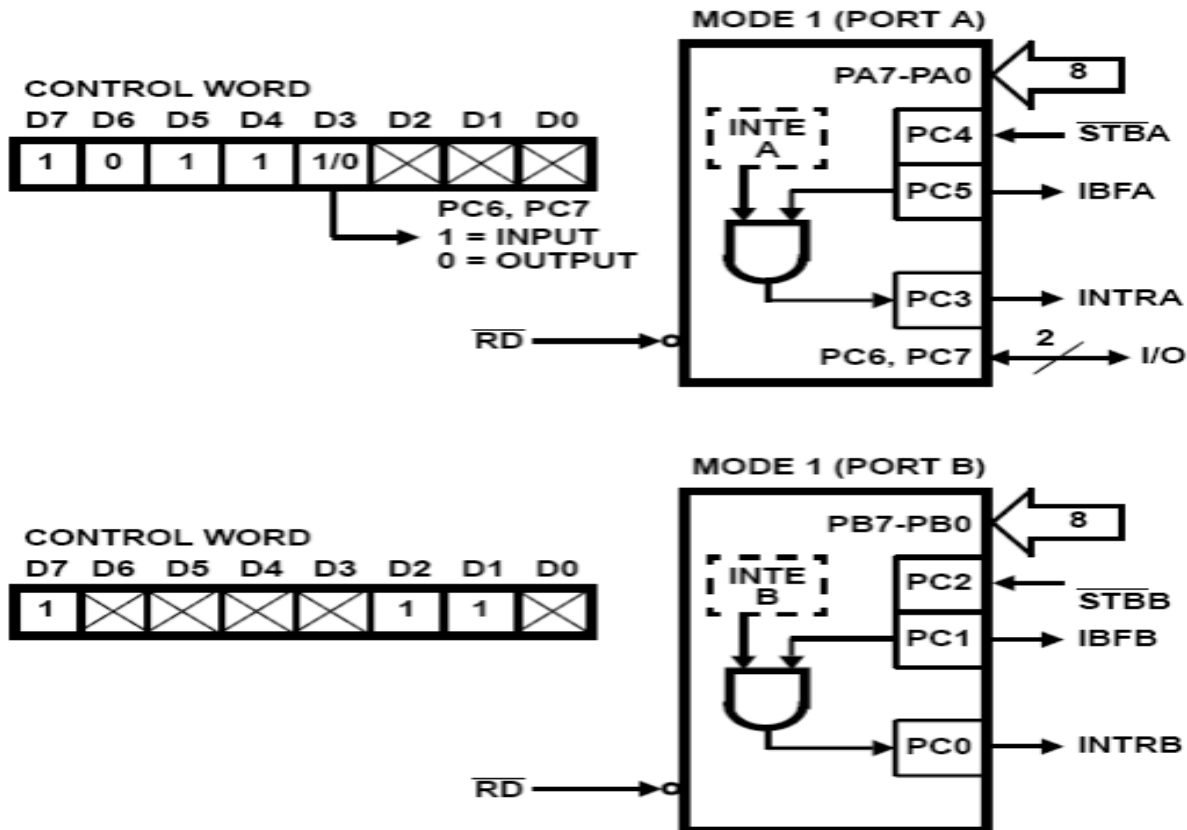
```
MOV AL,083H  
OUT PPICR , AL  
IN AL, PPIB  
OUT PPIA, AL  
IN AL, PPIC  
AND AL, 0FH  
MOV CL, 04H  
ROL AL, CL  
OUT PPIC, AL  
HLT
```

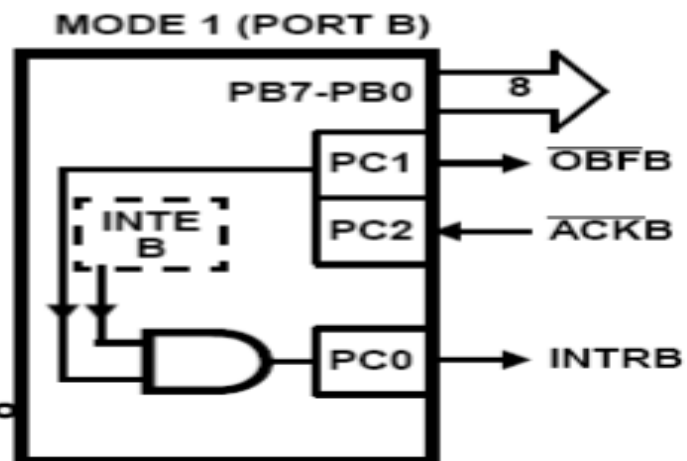
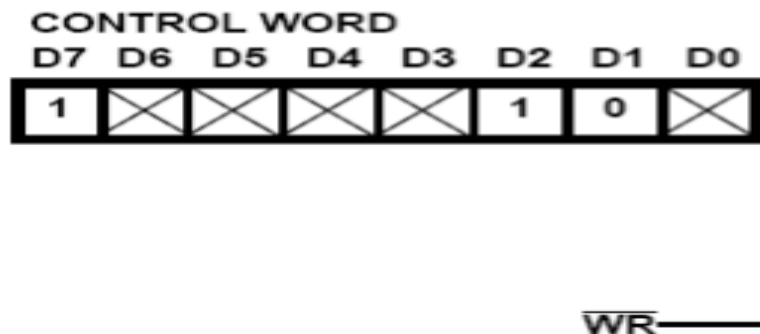
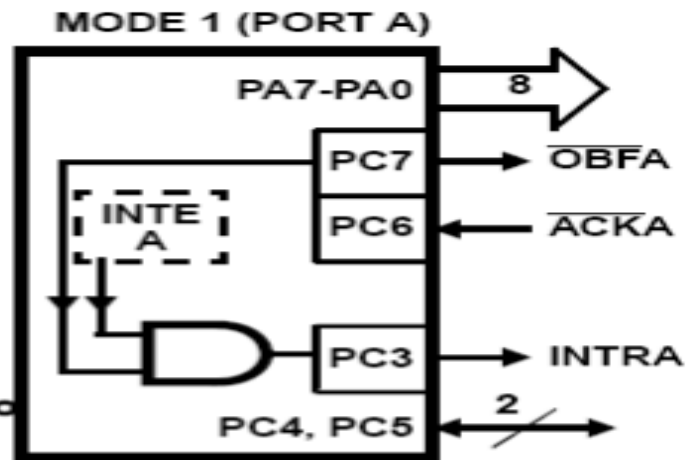
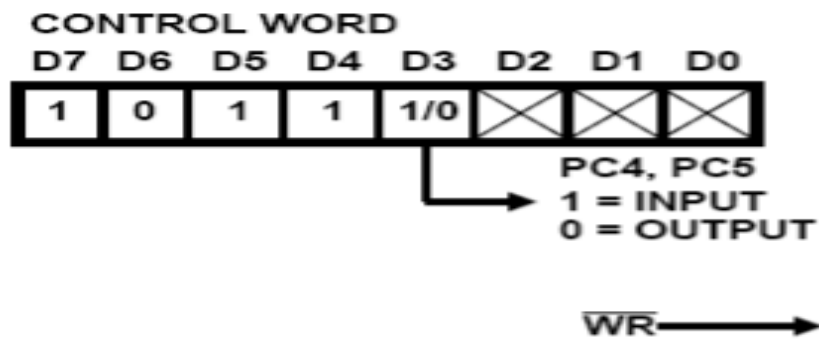

Interrupt Enable in Mode 1

INTE (Interrupt Enable): This is an internal flip flop used to enable or disable the generation of the INTR signal. The two flip flops are set/reset using the BSR mode.

For Input Mode : INTEa → PC4 and INTEb → PC2

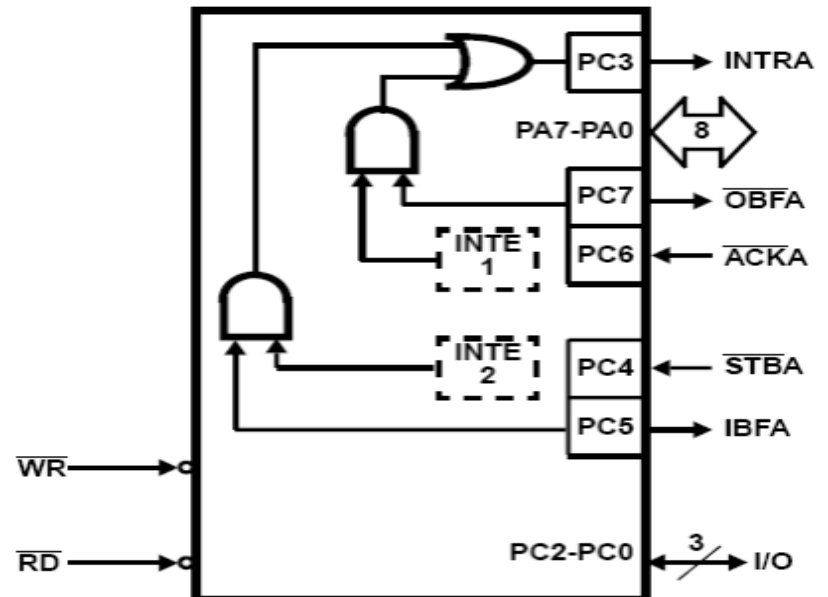
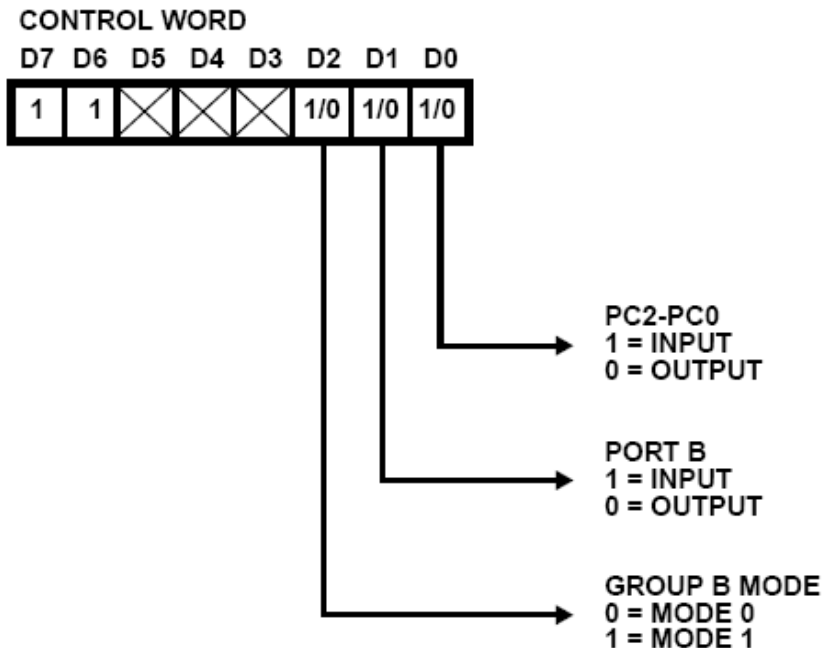
For Output Mode: INTEa → PC6 and INTEb → PC2





Interrupt Enable in Mode 2

INTE (Interrupt Enable): INTE1 and INTE2 enables the INTR pin. The state of the INTR pin is controlled through PORT C bits PC6 (INTE1) and PC4 (INTE2)



- Interface an 8255 chip with 8086 to work as an I/O port. Initialize port A as output port, Port B as I/P port and Port C as O/P port. Port A address should be 0740H. Write an ALP to sense switch positions SW0–SW7 connected at port B. The sensed pattern is to be displayed on port A, to which 8 LED's are connected, while port C lower displays number of on switches out of the total eight switches ?



Control Word

B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	1	0

Control Word=82h

Port A=0740

Port B=0742

Port C=0744

CWR=0746

MOV DX,0746H

MOV AL, 82H

OUT DX, AL

SUB DX, 04

IN AL,DX

SUB DX, 02;

OUT DX,AL

MOV BL, 00H;

MOV CH,08H;

Label1 : ROL AL,1

JNC label 2

INC BL

Label2: DEC CH

JNZ Label1

MOV AL,BL

ADD DX,04

OUT DX,AL

HLT